

Course: Cloud and Network Security CNS-C2-2026

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Week 4 Assignment 2:

Packet Tracer WLAN configuration

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WLAN Configuration table

Packet Tracer - WLAN Configuration

Addressing Table

Device	Interface	IP Address
Home Wireless Router	Internet	DHCP
	LAN	192.168.8.1/27
RTR-1	G0/0/0.2	192.168.2.1/24
	G0/0/0.5	192.168.5.1/24
	G0/0/0.100	192.168.100.1/24
	G0/0/1	10.6.0.1/24
SW1	VLAN 200	192.168.100.100/24
LAP-1	G0	DHCP
WLC-1	Management	192.168.100.254/24
RADIUS Server	NIC	10.6.0.254/24
Home Admin	NIC	DHCP
Enterprise Admin	NIC	192.168.100.200/24
Web Server	NIC	203.0.113.78/24
DNS Server	NIC	10.100.100.252
Laptop	NIC	DHCP
Tablet PC	Wireless0	DHCP
Smartphone	Wireless0	DHCP
Wireless Host 1	Wireless0	DHCP
Wireless Host 2	Wireless0	DHCP

WLAN Information

WLAN	SSID	Authentication	Username	Password
Home Network	HomeSSID	WPA2-Personal	N/A	Cisco123
WLAN VLAN 2	SSID-2	WPA2 Personal	N/A	Cisco123
WLAN VLAN 5	SSID-5	WPA2 Enterprise	userWLAN5	userV5pass

Note: It is not a good practice to reuse passwords as is done in this activity. Passwords have been reused to make it easier to work through the tasks.

Objectives

In this activity, you will configure both a wireless home router and a WLC-based network. You will implement both WPA2-PSK and WPA2-Enterprise security.

Configure a home router to provide Wi-Fi connectivity to a variety of devices.

Configure WPA2-PSK security on a home router.

Configure interfaces on a WLC. *(highlighted)*

Configure WLANs on a WLC. *(highlighted)*

Configure WPA2-PSK security on a WLAN and connect hosts to WLAN.

Configure WPA2-Enterprise on a WLAN and connect hosts to the WLAN.

Verify connectivity WLAN connectivity.

Introduction

This lab covered two wireless setups: a home router and an enterprise network built around a Wireless LAN Controller (WLC). In Part 1, I configured a home wireless router with custom DHCP settings, a 2.4 GHz SSID on channel 6, WPA2-Personal security, and a changed admin password. In Part 2, I configured a WLC with two VLAN interfaces, an internal DHCP scope for management, and RADIUS and SNMP server settings. I then created two WLANs, one using WPA2-PSK and one using WPA2-Enterprise (802.1X), both with FlexConnect local

switching and local authentication. Wireless hosts were connected to each network to verify the configurations.

Steps

Part 1: Configure a Home Wireless Router.

Step 1: Change DHCP settings.

Step 2: Configure the Wireless LAN.

Step 3: Configure security.

Step 4: Connect clients to the network. *(continued from previous page)*

Part 2: Configure a WLC Controller Network

Step 1: Configure VLAN interfaces.

Step 2: Configure a DHCP scope for the wireless management network.

Step 3: Configure the WLC with external server addresses.

Step 4: Create the WLANs.

Step 5: Configure the hosts to connect to the WLANs.

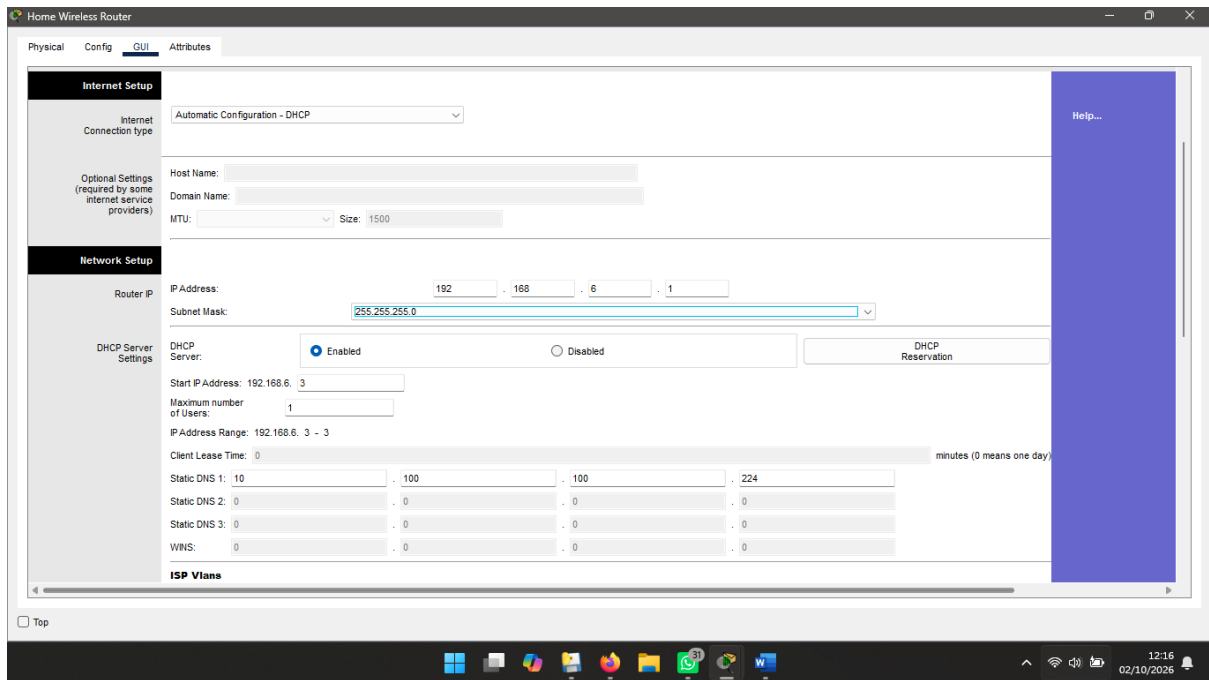
Step 6: Test connectivity.

Part 1: Configure a Home Wireless Router.

Step 1: Change DHCP settings.

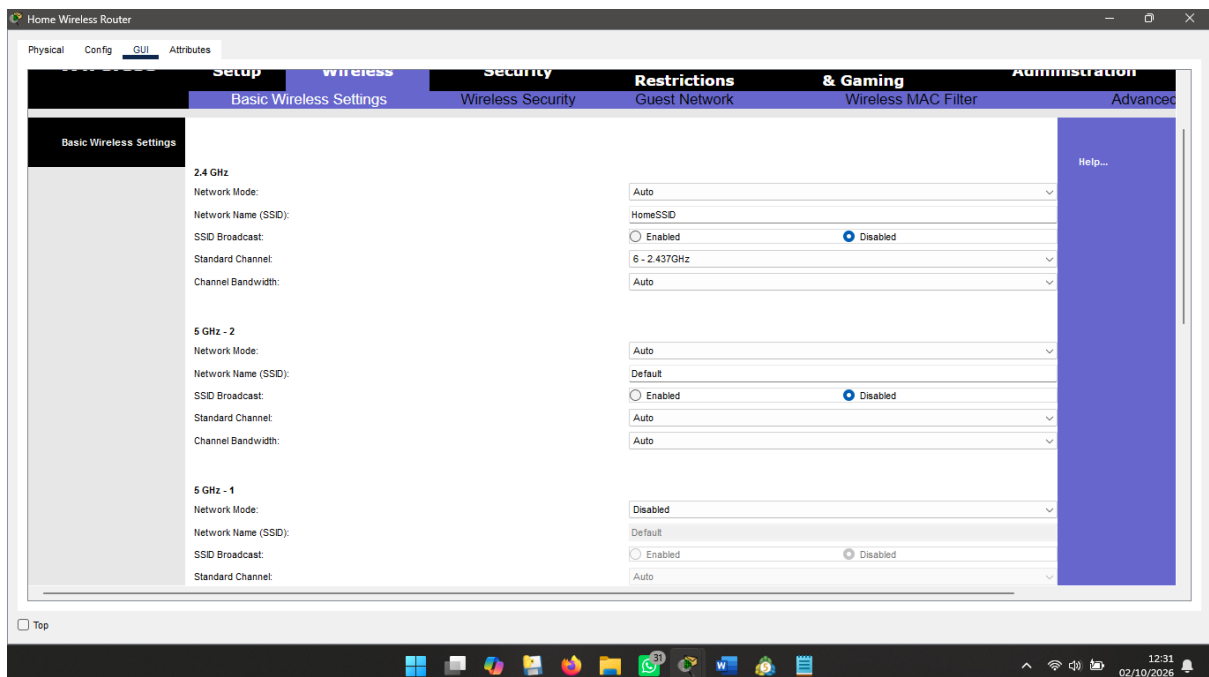
Open the Home Wireless Router GUI and change the router IP and DHCP settings according to the information in the Addressing Table. Permit a maximum of 20 addresses to be issued by the router. Configure the DHCP server to start with IP address .3 of the LAN network.

Configure the internet interface of the router to receive its IP address over DHCP. Configure the static DNS server to the address in the Addressing Table.



Step 2: Configure the Wireless LAN.

The network will use the 2.4GHz Wireless LAN interface. Configure the interface with the SSID shown in the Wireless LAN information table. Use channel 6. Be sure that all wireless hosts in the home will be able to see the SSID.



Step 3: Configure security.

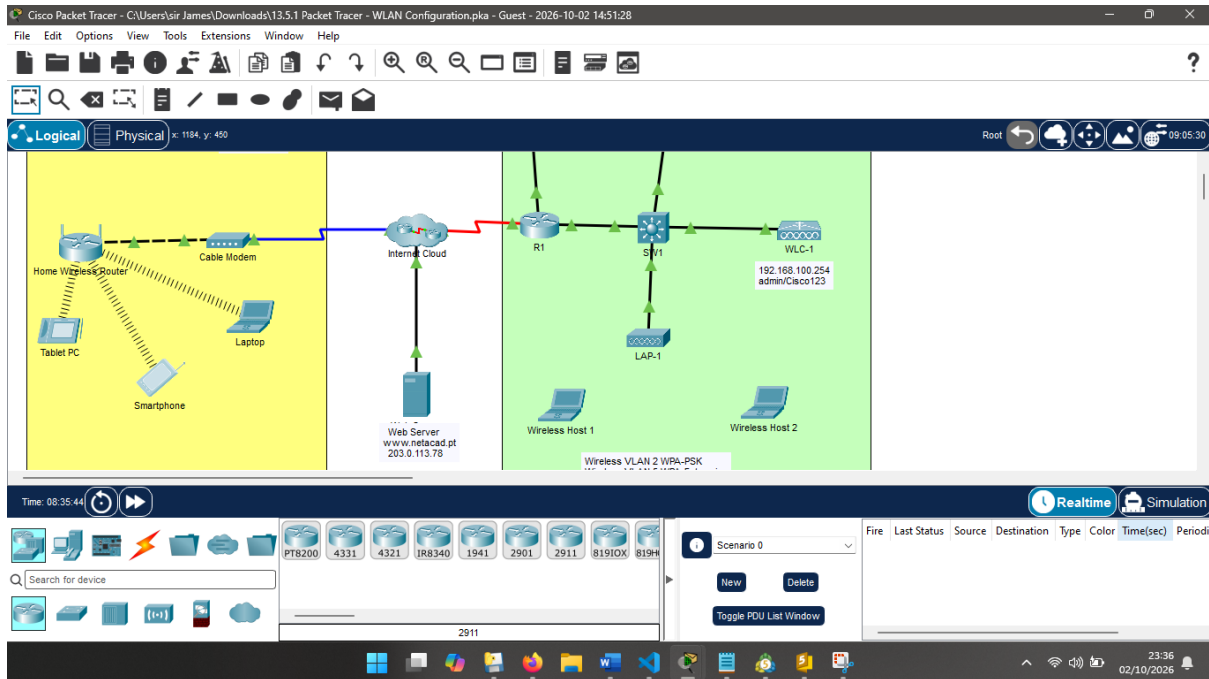
Configure wireless LAN security. Use WPA2 Personal and the passphrase shown in the Wireless LAN information table. Secure the router by changing the default password to the value shown in the Wireless LAN information table.

The screenshot shows the 'Security' configuration page of a 'Wireless-N Broadband Router WRT300N' with firmware version v0.9.7. The page is divided into tabs: Physical, Config, GUI (selected), and Attributes. Under the 'Config' tab, there are sub-tabs: Setup, Wireless, Security (selected), Access Restrictions, Applications & Gaming, Administration, and Status. The 'Security' sub-tab is further divided into: Basic Wireless Settings, Wireless Security (selected), Guest Network, Wireless MAC Filter, and Advanced Wireless Settings. The 'Wireless Security' section shows settings for three wireless networks: 2.4 GHz, 5 GHz - 1, and 5 GHz - 2. For the 2.4 GHz network, the Security Mode is set to 'WPA2 Personal', Encryption is 'AES', and the Passphrase is 'Cisco123'. The Key Renewal is set to 3600 seconds. The 5 GHz - 1 and 5 GHz - 2 networks have their Security Modes set to 'Disabled'. A 'Help...' link is visible on the right side of the page.

The screenshot shows the 'Administration' configuration page of the same router. The 'Administration' sub-tab is selected, showing options for Management, Log, Diagnostics, Factory Defaults, and Firmware Upgrade. The 'Management' section is expanded, showing settings for Router Access, Web Access, and Remote Access. The Router Access section has fields for Router Password and Re-enter to confirm, both masked with dots. The Web Access section has checkboxes for Web Utility Access (HTTP/HTTPS) and Web Utility Access via Wireless (Enabled/Disabled). The Remote Access section has checkboxes for Remote Management (Enabled/Disabled), Web Utility Access (HTTP/HTTPS), and Remote Upgrade (Enabled/Disabled). There is also a section for Allowed Remote Ip Address with radio buttons for 'Any Ip Address' and a specific IP address range. The Remote Management Port is set to 8080. A 'Help...' link is visible on the right side of the page.

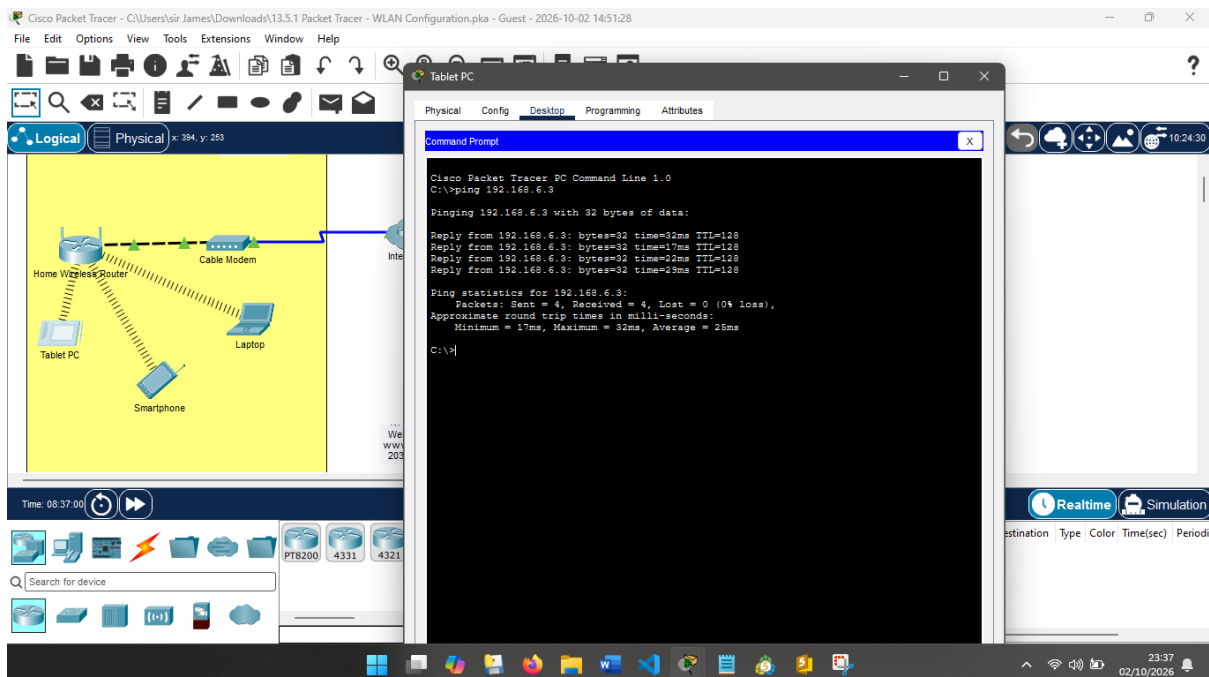
Step 4: Connect clients to the network.

Open the PC Wireless app on the desktop of the laptop and configure the client to connect to the network. Open the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network. Verify connectivity. The hosts should be able to ping each other and the web server. They should also be able to reach the web server URL.



Ping test

Ping laptop from tablet



Ping tablet from smartphone

The screenshot displays the Cisco Packet Tracer interface with a network topology and a command prompt window.

Network Topology:

- A **Home Wireless Router** is connected to a **Cable Modem** via a dashed line.
- The **Home Wireless Router** is connected to three devices: a **Tablet PC**, a **Laptop**, and a **Smartphone**.

Command Prompt Window:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.6.1

Pinging 192.168.6.1 with 32 bytes of data:

Reply from 192.168.6.1: bytes=32 time=39ms TTL=255
Reply from 192.168.6.1: bytes=32 time=91ms TTL=255
Reply from 192.168.6.1: bytes=32 time=17ms TTL=255
Reply from 192.168.6.1: bytes=32 time=15ms TTL=255

Ping statistics for 192.168.6.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 15ms, Maximum = 39ms, Average = 28ms

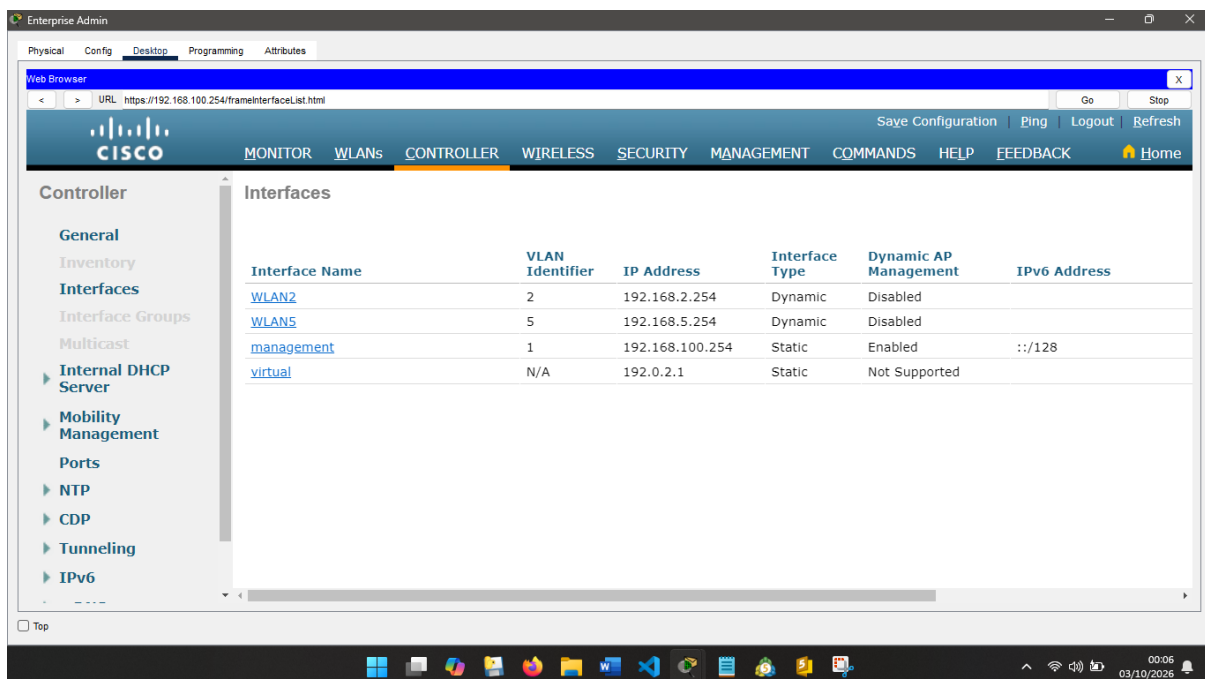
C:\>
```

The interface also shows a **Logical** view of the network, a **Physical** view, and a **Realtime** simulation mode. The status bar at the bottom indicates the time is 23:38 on 02/10/2026.

Part 2: Configure a WLC Controller Network

Step 1: Configure VLAN interfaces.

From the Enterprise Admin, navigate to the WLC-1 management interface via a web browser. To log into WLC-1, use admin as the username and Cisco123 as the password. Configure an interface for the first WLAN Name: WLAN 2, VLAN Identifier: 2, Port Number: 1, Interface IP Address: 192.168.2.254, Netmask: 255.255.255.0, Gateway: RTR-1 G0/0/0.2 address, Primary DHCP Server: Gateway address. Configure an interface for the second WLAN. Name: WLAN 5, VLAN Identifier: 5, Port Number: 1, Interface IP Address: 192.168.5.254, Netmask: 255.255.255.0, Gateway: RTR-1 interface G0/0/0.5 address, Primary DHCP Server: Gateway address.

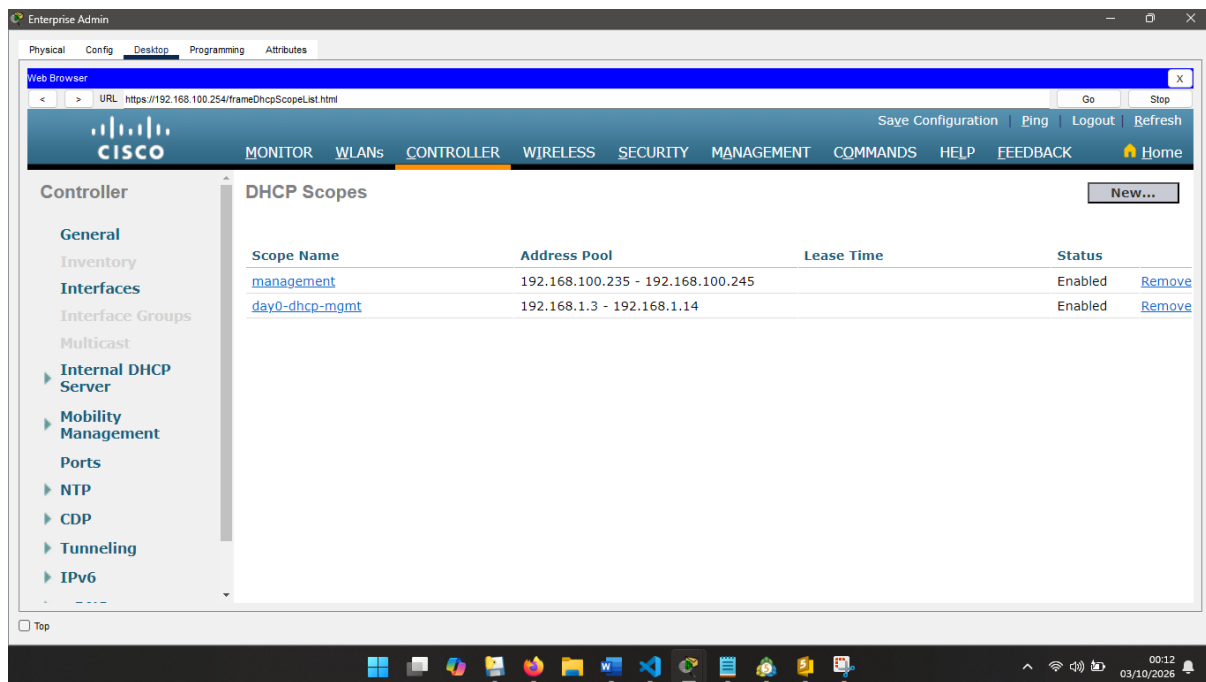


The screenshot shows the Cisco Enterprise Admin web interface. The 'CONTROLLER' tab is selected, and the 'Interfaces' section is displayed. The table below shows the configured interfaces:

Interface Name	VLAN Identifier	IP Address	Interface Type	Dynamic AP Management	IPv6 Address
WLAN2	2	192.168.2.254	Dynamic	Disabled	
WLAN5	5	192.168.5.254	Dynamic	Disabled	
management	1	192.168.100.254	Static	Enabled	::/128
virtual	N/A	192.0.2.1	Static	Not Supported	

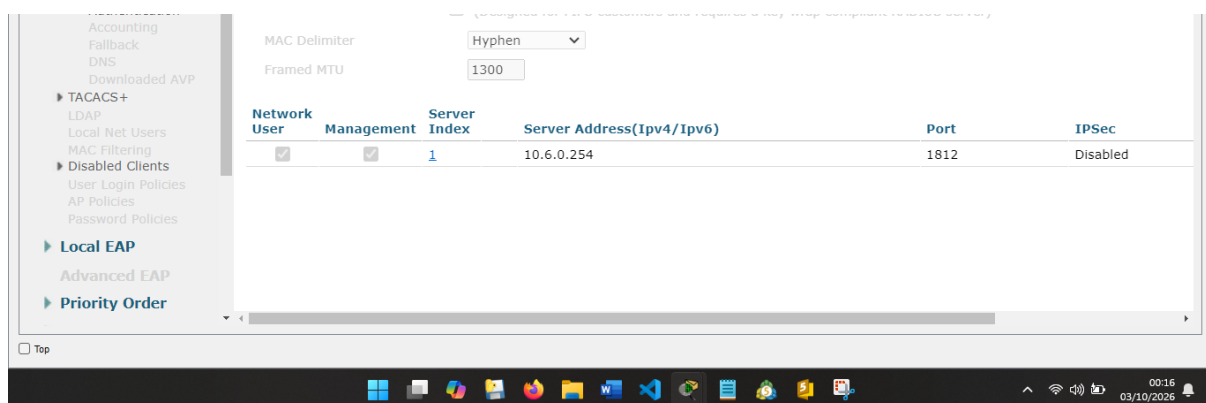
Step 2: Configure a DHCP scope for the wireless management network.

Configure and enable an internal DHCP scope as follows: Scope Name: management, Pool Start Address: 192.168.100.235, Pool End Address: 192.168.100.245, Network: 192.168.100.0, Netmask: 255.255.255.0, Default Routers: 192.168.100.1



Step 3: Configure the WLC with external server addresses.

Configure the RADIUS server information as follows: Server Index: 1, Server Address: 10.6.0.254, Shared Secret: RadiusPW.



Configure the WLC to send logs information to an SNMP server. Community Name: WLAN, IP Address: 10.6.0.254

The screenshot shows the WLC configuration interface. On the left, a sidebar menu includes 'Summary', 'SNMP', 'HTTP-HTTPS', 'IPSEC', 'Telnet-SSH', 'Serial Port', 'Local Management Users', 'User Sessions', 'Logs', 'Mgmt Via Wireless', 'Cloud Services', 'Software Activation', and 'Tech Support'. The 'SNMP' section is selected, displaying a table with the following data:

SNMP Trap Receiver Name	IP Address(Ipv4/Ipv6)	Status	IPSec
WLAN	10.6.0.254	Enable	Disable

At the bottom of the page, there is a 'Top' link and a Windows taskbar with various application icons and a system clock showing 00:19 on 03/10/2026.

Step 4: Create the WLANs.

Create the first WLAN: Profile Name: Wireless VLAN 2, WLAN SSID: SSID-2, ID: 2, Interface: WLAN 2, Security: WPA2-PSK, Passphrase: Cisco123. Under the Advanced tab, go to the FlexConnect section. Enable FlexConnect Local Switching and FlexConnect Local Auth.

The screenshot shows the WLC configuration interface with the 'Advanced' tab selected. The left sidebar menu includes 'Advanced' and 'AP Groups'. The main content area displays a table with the following data:

WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies
2	WLAN	Wireless VLAN 2	SSID-2	Disabled	[WPA2][Auth(PSK

At the bottom of the page, there is a 'Top' link and a Windows taskbar with various application icons and a system clock showing 00:29 on 03/10/2026.

Create the second WLAN: Profile Name: Wireless VLAN 5, WLAN SSID: SSID-5, Interface: WLAN 5, ID: 5, Security: 802.1x - WPA2-Enterprise

Configure the WLAN to use the RADIUS server for authentication. Make the FlexConnect settings as was done in Step 4a.

☐	WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policy
☐	2	WLAN	Wireless VLAN 2	SSID-2	Disabled	[WPA2][Auth(PSK
☐	5	WLAN	Wireless VLAN 5	SSID-5	Disabled	[WPA2][Auth(802

Step 5: Configure the hosts to connect to the WLANs.

Use the desktop PC Wireless app to configure the hosts as follows: Wireless Host 1 should connect to Wireless VLAN 2. Wireless Host 2 should connect to Wireless VLAN 5 using the credentials in the WLAN information table.

The screenshot displays the Cisco Packet Tracer interface. On the left, a network diagram shows a central switch (SW1) connected to a router (R1) and a laptop (LAP-1). Two wireless hosts, Wireless Host 1 and Wireless Host 2, are connected to the switch. A web server is connected to the router. The switch is configured with two WLANs: Wireless VLAN 2 WPA-PSK and Wireless VLAN 5 WPA-Enterprise. On the right, two windows show the configuration for Wireless Host 1 and Wireless Host 2. Both hosts are configured to connect to the 2.4GHz network. Wireless Host 1 is configured to connect to the Wireless Network Name SSID-2, and Wireless Host 2 is configured to connect to the Wireless Network Name SSID-5. Both hosts are configured with the same IP address (192.168.2.2), Subnet Mask (255.255.255.0), Default Gateway (10.100.100.252), and DNS1 (10.100.100.252). The security policy for both hosts is WPA2-Enterprise.

Step 6: Test connectivity.

Test connectivity between the wireless hosts and the web server by ping and URL.

Ping host 2 from host 1

The network diagram shows a topology with an Internet Cloud connected to a Web Server (www.netacad.pt, 203.0.113.78). A central router R1 is connected to a switch S1. S1 is connected to a Wireless LAN Controller (WLC-1) at 192.168.100.254. S1 also has a LAN interface connected to a Laptop (LAP-1). S1 is configured with Wireless VLAN 2 WPA-PSK and Wireless VLAN 5 WPA-Enterprise. Two wireless hosts, Wireless Host 1 and Wireless Host 2, are connected to S1. Wireless Host 1 is configured with IP 192.168.5.2 and subnet mask 255.255.255.0. Wireless Host 2 is configured with IP 192.168.5.1 and subnet mask 255.255.255.0. Both hosts have a default gateway of 192.168.5.1.

Terminal output for Wireless Host 1:

```
Wireless Host 1
Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.5.2: bytes=32 time=32ms TTL=127
Reply from 192.168.5.2: bytes=32 time=16ms TTL=127
Reply from 192.168.5.2: bytes=32 time=31ms TTL=127

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 32ms, Average = 26ms

C:\>
```

Terminal output for Wireless Host 2:

```
Wireless Host 2
Wireless0 Connection: (default port)
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: FE80::20C:85FF:FE9B:9E35
IPv6 Address...: ::
IPv4 Address...: 192.168.5.2
Subnet Mask...: 255.255.255.0
Default Gateway...: 192.168.5.1

Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>
```

Ping radius server from wireless host 2

The network diagram is the same as the previous one, but with the addition of a RADIUS Server configuration window. The RADIUS Server is configured with the following settings:

- Display Name: RADIUS Server
- Gateway/DNS IPv4: Static, Default Gateway: 10.6.0.1, DNS Server: (empty)
- Gateway/DNS IPv6: Automatic, Default Gateway: (empty), DNS Server: (empty)
- Device Clock: 07:12:28 Fri Sep 6 2019 UTC
- PTP: (empty)
- Profile: (empty)
- Enable/Disable: Disable

Terminal output for Wireless Host 1 (same as above):

```
Wireless Host 1
Bluetooth Connection:
Connection-specific DNS Suffix...:
Link-local IPv6 Address...: ::
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 192.168.5.2

Pinging 192.168.5.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.5.2: bytes=32 time=32ms TTL=127
Reply from 192.168.5.2: bytes=32 time=16ms TTL=127
Reply from 192.168.5.2: bytes=32 time=31ms TTL=127

Ping statistics for 192.168.5.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 32ms, Average = 26ms

C:\>
```

Terminal output for Wireless Host 2:

```
Wireless Host 2
IPv6 Address...: ::
IPv4 Address...: 0.0.0.0
Subnet Mask...: 0.0.0.0
Default Gateway...: 0.0.0.0

C:\>ping 10.6.0.1

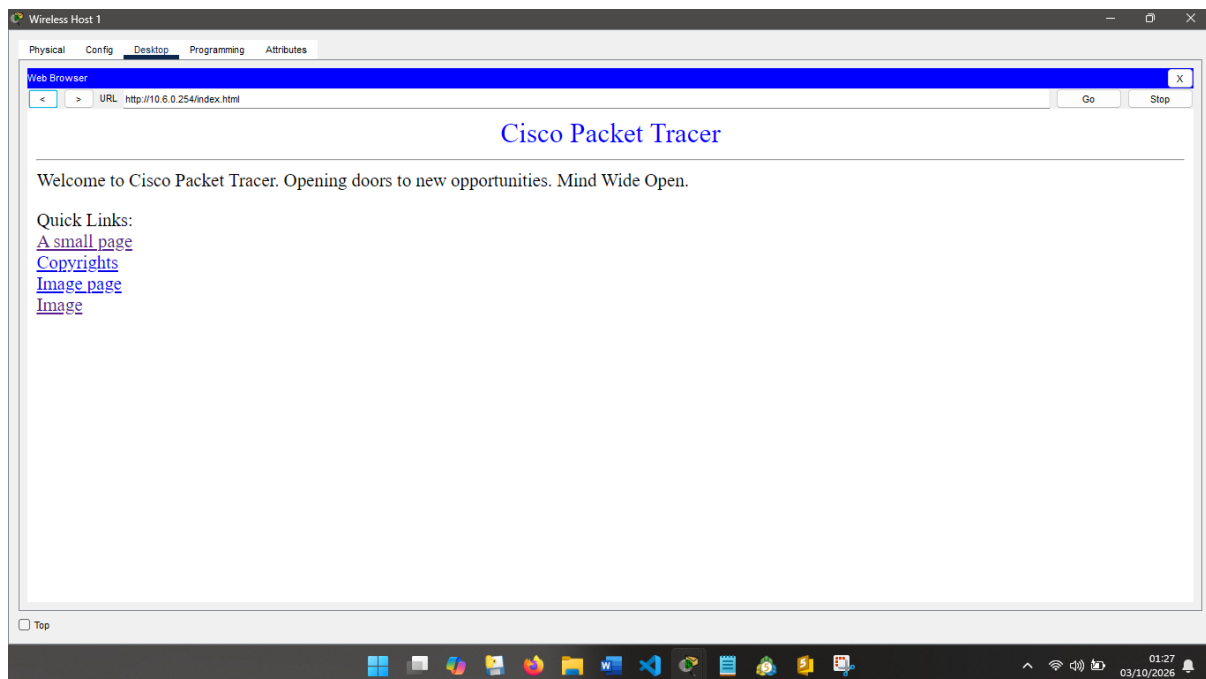
Pinging 10.6.0.1 with 32 bytes of data:

Reply from 10.6.0.1: bytes=32 time=16ms TTL=255
Reply from 10.6.0.1: bytes=32 time=16ms TTL=255
Reply from 10.6.0.1: bytes=32 time=12ms TTL=255
Reply from 10.6.0.1: bytes=32 time=11ms TTL=255

Ping statistics for 10.6.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 16ms, Average = 10ms

C:\>
```

Running server url on host 1



Conclusion

The lab showed the difference between personal and enterprise wireless security. WPA2-PSK relies on one shared passphrase, which is simple to set up but offers no per-user control. WPA2-Enterprise authenticates each user through a RADIUS server, which is more secure and scalable but needs more configuration, including AAA server settings and correct WLAN security parameters. After setup, both wireless hosts received IP addresses from the correct subnets and could ping and browse to the web server. This confirmed that the router, WLC interfaces, DHCP scopes, WLANs, and RADIUS authentication were all working together correctly. The lab also reinforced that most wireless problems come from small mismatches such as passphrases, shared secrets, interface mappings, or DHCP settings.